

Vibha Keremane Vasanth

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Education

University of Florida

Master of Science in Electrical and Computer Engineering

Gainesville, FL

May 2021

BMS Institute of Technology

Bachelor of Engineering in Telecommunication Engineering

Bengaluru, India

June 2019

Relevant Coursework

Neural Networks and Deep Learning, Machine Learning, Image Processing and Computer Vision, Computational Photography, Digital Signal Processing, Advanced Data Structures, VLSI.

Technical Skills

Languages: Python, C++, MATLAB, Verilog, SQL

Tools: LabVIEW, HFSS, Cadence

Libraries: Pandas, NumPy, Matplotlib, Tensorflow, PyTorch

Projects

Implementation of QMI / University of Florida, FL, USA

Nov 2020 - Dec 2020

- Developed two models, one using autoencoder for preprocessing and CNN (Convolutional Neural Networks) and MLP (Multilayer Perceptron) architectures for classification. The second model used QMI (Quadratic Mutual Information) loss function and Maximum a posteriori classifier.
- Compared the two model architectures for Fashion MNIST dataset for criteria such as learning rate, kernel size, number of trainable parameters and accuracy.

Evaluation of Deep Learning Architectures / University of Florida, FL, USA

Nov 2020 - Dec 2020

- Developed and compared the performance of MLP (Multilayer Perceptron) and CNN (Convolutional Neural Networks) classifiers on Fashion MNIST dataset.
- Conducted experiments to exhaustively study the effect of different hyperparameters, training procedures on the performance of the networks and compared the results using confusion matrices.

Hashtag Counter / University of Florida, FL, USA

Mar 2020 - Apr 2020

- Implemented a system to find the most popular hashtags on social media and the frequency of the hashtags using C++. The most popular hashtag was extracted.
- Exercised Fibonacci Heap to implement a max priority queue based on the frequency of hashtags and Hash Table to keep track of hashtags in the heap.

Line and Circle Detection using Hough Transform / University of Florida, FL, USA

Nov 2019 - Dec 2019

- Identified lines and circles in an image by finding the corresponding peaks in Hough Space.
- Detected lines in an image by joining the edges using Canny edge detection.

Handwritten Character Recognition using Neural Nets / University of Florida, FL, USA

Nov 2019 - Dec 2019

- Developed a program for classification/identification of handwritten characters given an image of the handwritten character as an input using the gradient descent approach.
- Obtained a high level of classification accuracy on a data set which consisted of eight characters using a model which requires minimal pre-processing.

Human Assisting Robot / BMS Institute of Technology, India

Sep 2018 - May 2019

- Designed and fabricated a robot that tracks the target using Bluetooth and moves towards it while doing the tracking through GPS.

Certifications

DeepLearning.ai / Courseera.org

Oct 2020 - Mar 2021

- Acquired knowledge of Deep Learning techniques from courses like Convolutional Neural Networks, Structuring Machine Learning Projects, Sequence Models etc.
- Developed and trained models through projects like Classification of SIGNS dataset using Tensorflow.